

Technical Datasheet

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****Product**:** Long-endurance drone battery cell (design V4 Thermal-tuned) | ****Date**:** 2026-08-25

****System**:** NMC811 / graphite, high-transport electrolyte (t_{ion}=0.50 estimate)

Field	Value	Source
Rated capacity	3.40 Ah (nominal, design) / 3.45 Ah (1C simulation-verified)	params_r1_v4.json / cell/r1_v4_1c.json
Nominal voltage / window	3.91 V (midpoint) / 4.20–2.50 V	cell/r1_v4_energy.json / parameter set
Rated energy	12.68 Wh (∫V·I dt, simulation)	cell/r1_v4_energy.json
Energy density	473.0 Wh/kg (contract caliber, electrolyte/casing excluded)	cell/r1_v4_energy.json
Volumetric energy density	912.2 Wh/L	cell/r1_v4_energy.json
Max continuous discharge rate	5C verified: retention 99.0 %, T_max 48.1 °C	cell/r1_v4_5c.json
Fast-charge capability	4C@45°C: T_max 331.98 K (≤333.15), no plating (min anode 0.0247 V)	cell/r1_v4_4c.json
DC resistance (1C)	1.38 mΩ	cell/r1_v4_energy.json
Peak power density (theoretical)	114.8 kW/kg	cell/r1_v4_energy.json
Operating temperature range	–20...45 °C (simulation conditions used: 25 °C discharge, 45 °C charge; –20 °C not verified)	honest note
Cycle life	Indicative only: 100-cycle aging → SEI 467.2 nm; capacity trajectory shows standard-SEI-model artifact (climb-then-saturate) — treat as not reliably simulated	cell/r1_v4_aging.json
Safety determination	4C charge: no plating, T_max within limit; overcharge to 4.7 V: T_max 300.57 K, thermal runaway not triggered	cell/r1_v4_4c.json, r1_v4_tr.json
Dimensions	Electrode 65 mm × 1580 mm; stack thickness 135.3 μm (excl. casing); casing dims not provided	parameter set
Mass	26.8 g (contract stack caliber); full BOM incl. electrolyte ≈ 31.6 g	cell/r1_v4_energy.json / bom.xlsx